Calculus & Optimization

Revision companion

Use this after the guided lessons: cover each worked answer, recall the rule, then solve the example on paper. Every module includes its key formulas, conditions and common traps. Full explanations and runnable examples are in the companion HTML notes.

Scope: Every named Section 3 topic. Integration, gradients and Hessians form an additional supporting bridge.

Official GATE DA 2027 syllabus ↗

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01 / Functions & algebra refresh

Domains, graphs, composition, exponentials, logarithms and trigonometry

Functions, domains and graphs

A real function assigns exactly one output to each input in its domain. The domain is the allowed input set, the codomain is the declared target set, and the range is the set actually attained. A graph plots pairs (x,f(x)); a vertical line can meet it at most once for each input.

Domain of $\sqrt{(x-1)}/(x-3)$: $[1,3) \cup (3,\infty)$

Work it through: For $f(x)=(x^2-1)/(x-1)$, $x=1$ is excluded. For $x \neq 1$, factor and cancel to obtain $f(x)=x+1$. The graph is the line $y=x+1$ with a hole at (1,2). Defining $f(1)=2$ would create a new, continuous extension.

Watch the trap: A simplified expression can have a larger domain than the original function.

Recall task: Find the real domain of $\ln(x-2)/\sqrt{(5-x)}$.

Worked answer: $x>2$ for the logarithm and $x<5$ because the square root is in the denominator. Domain: (2,5).

Composition, inverses and algebraic preparation

The composition $(f \circ g)(x)=f(g(x))$ requires x in the domain of g and g(x) in the domain of f. Order matters. An inverse function reverses input and output, and exists on the full range when f is one-to-one. Restrict a domain when needed: x^2 is invertible on $[0,\infty)$, not on all real numbers.

$(f \circ g)(x)=f(g(x)) \cdot f^{-1}(f(x))=x$ on an invertible domain

Work it through: Let $f(t)=\sqrt{t}$ and $g(x)=x-2$. $f(g(x))=\sqrt{(x-2)}$, with domain $x \geq 2$. The reverse composition is $g(f(x))=\sqrt{x-2}$, with domain $x \geq 0$. For $h(x)=3x+2$, solve $y=3x+2$ to get $h^{-1}(y)=(y-2)/3$.

Watch the trap: $f^{-1}(x)$ denotes inverse function, not $1/f(x)$.

Recall task: Find the inverse of $f(x)=x^2+1$ on $x \geq 0$ and its domain.

Worked answer: $f^{-1}(y)=\sqrt{(y-1)}$, with $y \geq 1$.

Exponential, logarithmic and trigonometric essentials

For $a>0$ and $a \neq 1$, a^x is positive and has inverse $\log_a x$. The natural exponential e^x and natural logarithm $\ln x$ are especially convenient for calculus. Laws include $\ln(ab)=\ln a+\ln b$ and $\ln(a^r)=r \ln a$ for $a>0$; there is no corresponding sum law $\ln(a+b)=\ln a+\ln b$.

$\ln(ab)=\ln a+\ln b \cdot a^x=e^{x \ln a} \cdot 180°=\pi$ radians

Work it through: $\ln(e^3)=3$ because ln and exp are inverse functions. For $x=0.1$ radians, $\sin x \approx 0.0998334$, close to x. $\cos x \approx 0.995004$, close to $1-x^2/2=0.995$. In degrees the same derivative formulas would require a $\pi/180$ factor.

Watch the trap: The standard sin and cos derivatives assume radians, and logarithm rules require positive real arguments.

Recall task: Simplify $\ln(e^2\sqrt{e})$ and compare $\sin(0.01)$ with 0.01.

Worked answer: The logarithm is $2+1/2=2.5$. $\sin(0.01) \approx 0.0099998333$.

02 / Limits & continuity

One-sided behavior, standard limits, discontinuities and existence theorems

What a limit really says

The statement $\lim_{x\to a} f(x)=L$ concerns values near a, with $x\neq a$. It does not require f(a) to exist or equal L. Formally, for every $\varepsilon>0$ there is $\delta>0$ so that $0<|x-a|<\delta$ implies $|f(x)-L|<\varepsilon$. You control the output error by restricting the input neighborhood.

$\lim_{x\to a} f(x)=L \Leftrightarrow$ f(x) approaches L from both sides of a

Work it through: For $(x^2-1)/(x-1)$, $x\neq 1$, cancel to obtain $x+1$. As x approaches 1 from either side, $x+1$ approaches 2. The limit is 2 even though the original expression is undefined at 1. For $|x|/x$, the left limit is -1 and right limit 1, so no two-sided limit exists.

Watch the trap: f(a), $\lim_{x\to a} f(x)$, and the existence of a limit are three separate questions.

Recall task: Find $\lim_{x\to 0} x^2 \cos(1/x)$.

Worked answer: It is 0 because $-x^2\le x^2\cos(1/x)\le x^2$ and both bounds tend to 0.

Standard limits and indeterminate forms

Memorize the foundational limits $\sin x/x\to 1$, $(1-\cos x)/x^2\to 1/2$, $(e^x-1)/x\to 1$ and $\ln(1+x)/x\to 1$ as $x\to 0$, with radians for trigonometry. Substitute a scaled variable carefully: $\sin(3x)/x\to 3$.

$\sin x/x\to 1 \cdot \ln(1+x)/x\to 1 \cdot (e^x-1)/x\to 1 \cdot (1+x)^{1/x}\to e$

Work it through: Evaluate $\lim_{x\to 0} (1-\cos(2x))/x^2$. Set $t=2x$, so $x^2=t^2/4$. The expression is $4(1-\cos t)/t^2$. Its limit is $4\times 1/2=2$.

Watch the trap: 0/0 is a signal to investigate; it is not zero and it is not automatically undefined as a limit.

Recall task: Find $\lim_{x\to 0} (e^{3x}-1)/\sin(2x)$.

Worked answer: Split into $(e^{3x}-1)/(3x) \times (3/2) \times (2x)/\sin(2x)$; the limit is 3/2.

Continuity and its consequences

Continuity at a means f(a) exists, the limit exists, and the limit equals f(a). At an included endpoint use the one-sided limit within the domain. Polynomials are continuous everywhere; rational functions are continuous where their denominators are nonzero.

Continuous at a $\Leftrightarrow \lim_{x\to a} f(x)=f(a)$ · continuous on [a,b] $\Rightarrow$ extrema attained

Work it through: Let $f(x)=kx+1$ for $x<2$ and $f(x)=x^2-1$ for $x\ge 2$. The right value at 2 is 3. Continuity requires $2k+1=3$. Thus $k=1$; this only establishes continuity, not matching derivatives.

Watch the trap: A continuous function may have corners, and a root guaranteed by a sign change need not be unique.

Recall task: Choose c so $f(x)=\sin x/x$ for $x\neq 0$ and $f(0)=c$ is continuous.

Worked answer: $c=1$ by the standard limit.

03 / Differentiation from scratch

Difference quotients, derivative rules, chain rule and higher derivatives

Derivative as a limiting slope

The derivative $f'(a)$ is $\lim_{h\to 0}[f(a+h)-f(a)]/h$ when the finite two-sided limit exists. It is the tangent slope and instantaneous rate of change. The numerator is an output change; h is the input change. The derivative has units of output per input.

$f'(a)=\lim_{h\to 0}\,[f(a+h)-f(a)]/h$ · tangent: $y=f(a)+f'(a)(x-a)$

Work it through: For $f(x)=x^2$ at $a=2$, expand $(2+h)^2-4=4h+h^2$. Divide by $h\neq 0$ to get $4+h$. The limit is 4, so $f'(2)=4$. The tangent line is $y=4+4(x-2)=4x-4$.

Watch the trap: The derivative is a limit, not the quotient with h=0 substituted directly.

Recall task: Use the definition to differentiate x^3 at $a=1$.

Worked answer: $[(1+h)^3-1]/h=3+3h+h^2\to 3$.

The derivative toolkit

For real powers where defined, $d(x^r)/dx=r\,x^{r-1}$. A constant differentiates to zero; sums and constant multiples differentiate term by term. The product rule is $(uv)'=u'v+uv'$. The quotient rule is $(u/v)'=(u'v-uv')/v^2$ where $v\neq 0$.

$(uv)'=u'v+uv'$ · $(u/v)'=(u'v-uv')/v^2$ · $(f\circ g)'=(f'\circ g)g'$

Work it through: Differentiate f(x)=x²e^{3x}. The derivative of x^2 is 2x; that of e^{3x} is 3e^{3x}. Product rule gives $f'=2xe^{3x}+3x^2e^{3x}$. Factor: $f'=e^{3x}(2x+3x^2)$.

Watch the trap: The derivative of a product is not the product of derivatives. Do not omit the inner derivative.

Recall task: Differentiate $\ln(1+x^2)$ and evaluate at $x=1$.

Worked answer: $f'(x)=2x/(1+x^2)$, so $f'(1)=1$.

Implicit, logarithmic and higher differentiation

Implicit differentiation treats y as a function of x. Differentiating $x^2+y^2=1$ gives $2x+2yy'=0$, so $y'=-x/y$ where $y\neq 0$. This formula is local on a branch, and fails to describe a finite slope where its denominator vanishes.

$y=u^v>0 \Rightarrow y'=u^v(v'\ln u+v\,u'/u)$ · $f^{(n)}$ = n-th derivative

Work it through: For $y=x^x$, take logs: $\ln y=x\ln x$. Differentiate: $y'/y=\ln x+1$. Multiply by y: $y'=x^x(\ln x+1)$. At $x=1$, $y'=1$.

Watch the trap: Applying the ordinary power rule to x^x treats the changing exponent as constant and gives the wrong result.

Recall task: Find the slope of $x^2+y^2=25$ at (3,4), and f'' for $f=x^4$.

Worked answer: The implicit slope is $-3/4$. For x^4, $f'=4x^3$ and $f''=12x^2$.

04 / What derivatives reveal

Mean value theorem, monotonicity, concavity and graph reasoning

Rolle’s theorem and the mean value theorem

Rolle’s theorem assumes continuity on [a,b], differentiability on (a,b), and equal endpoint values. It guarantees some c inside with f′(c)=0. It does not claim every stationary point has equal endpoints or that the guaranteed point is unique.

$$\exists c \in (a,b):\ f'(c)=[f(b)-f(a)]/(b-a)$$

Work it through: For $f=x^2$ on [1,3], the average slope is $(9-1)/(3-1)=4$. The function is continuous and differentiable on the required sets. Solve $2c=4$ to obtain $c=2$. The tangent at c is parallel to the secant through the endpoints.

Watch the trap: Rolle’s theorem does not apply to $|x|$ on $[-1,1]$ because differentiability fails at zero.

Recall task: Find the MVT point for $f=x^3$ on [0,2].

Worked answer: The average slope is 4; $3c^2=4$ gives $c=2/\sqrt{3}$, the positive point inside (0,2).

Sign charts, concavity and inflection

To determine increasing and decreasing intervals, find where f′ is zero or undefined within the domain, split the domain at those points, and test the derivative sign on each open piece. Positive means increasing; negative means decreasing. Keep holes and vertical asymptotes separate from critical points in the domain.

$$f'>0 \Rightarrow \text{increasing} \cdot f''>0 \Rightarrow \text{convex} \cdot \text{inflection requires concavity change}$$

Work it through: For $f=x^3-3x$, $f'=3(x^2-1)$, with zeros -1 and 1. The sign is +,−,+ on the three intervals. $f''=6x$ changes sign at zero, giving an inflection at (0,0). The local maximum is $f(-1)=2$ and local minimum $f(1)=-2$.

Watch the trap: A zero second derivative alone does not prove an inflection or classify a stationary point.

Recall task: Does x^4 have an inflection at zero? Explain with signs.

Worked answer: No. Its second derivative $12x^2$ is positive on both sides of zero, so concavity does not change.

Related rates and local error estimates

A related-rates problem links changing quantities through an equation, then differentiates with respect to time. If area $A=\pi r^2$ and radius depends on time, $dA/dt=2\pi r\, dr/dt$. Units are an excellent consistency check: length times length/time gives area/time.

$$\Delta f \approx f'(a)\Delta x \cdot |f(x)-L(x)| \le M|x-a|^2/2 \text{ when } |f''| \le M$$

Work it through: Approximate $\sqrt{4.04}$ using $a=4$. $f(4)=2$ and $f'(4)=1/4$. The change in x is 0.04, giving $L=2+0.25\times0.04=2.01$. The exact value is about 2.009975; the approximation is slightly high because $\sqrt{x}$ is concave.

Watch the trap: An approximation near a point can be poor far away, even if the derivative formula is exact.

Recall task: A circle has radius 3 cm increasing at 2 cm/s. Find its area growth rate.

Worked answer: $dA/dt=2\pi\times3\times2=12\pi$ cm²/s.

05 / Taylor series & approximation

Taylor polynomials, remainders, convergence and higher-order tests

Constructing a Taylor polynomial

A Taylor polynomial matches a function and its first n derivatives at a center a: $T_n(x)=\Sigma_{k=0}^{n} f^{(k)}(a)(x-a)^k/k!$. The constant term matches height, the linear term slope, the quadratic term curvature. Maclaurin means the special center a=0.

$$T_n(x)=\Sigma_{k=0}^{n} f^{(k)}(a)(x-a)^k/k!$$

Work it through: For e^x at a=0, every derivative at zero is 1. The degree-three polynomial is $1+x+x^2/2+x^3/6$. At x=0.2 it gives 1.2213333. The exact value is about 1.2214028, so the error is about 0.0000694.

Watch the trap: Do not omit factorial denominators or confuse polynomial degree with the number of nonzero terms.

Recall task: Write the degree-four Maclaurin polynomial for cos x and evaluate at 0.2.

Worked answer: $1-x^2/2+x^4/24$; at 0.2 it is approximately 0.9800666667.

Remainders and where a series works

Taylor's theorem with Lagrange remainder gives $f(x)=T_n(x)+f^{(n+1)}(\xi)(x-a)^{n+1}/(n+1)!$ for some ξ between a and x under the needed smoothness assumptions. If the derivative magnitude is bounded by M there, the error is at most $M|x-a|^{n+1}/(n+1)!$.

$$|Rn(x)|\leq M|x-a|^{n+1}/(n+1)!\cdot 1/(1-x)=\Sigma x^k \text{ for } |x|<1$$

Work it through: Approximate e^{0.1} by $1+x+x^2/2$. On [0,0.1], the third derivative e^x is at most e^{0.1}<1.106. Thus $|R_2|<1.106\times 0.1^3/6\approx 0.00018434$. The actual error is about 0.00017019, within the bound.

Watch the trap: A series identity has a domain of convergence; plugging in a value outside it is invalid.

Recall task: Bound the error of $\sin(0.1)\approx 0.1$ using a degree-two Taylor polynomial.

Worked answer: The degree-two polynomial is x; $|f'''|=|\cos x|\leq 1$, so error$\leq 0.1^3/6\approx 0.000166667$.

Leading terms, limits and flat stationary points

Near zero, the first nonzero Taylor term often determines a limit. Subtracting lower-order terms exposes the next term: e^x-1-x behaves like $x^2/2$. Keep enough terms to resolve all cancellations before dividing.

$$f(a+h)-f(a)\approx f^{(n)}(a)h^n/n! \text{ for the first nonzero derivative order } n$$

Work it through: Compute $\lim_{x\to 0} (e^x-1-x)/x^2$. Expansion gives $e^x=1+x+x^2/2+O(x^3)$. Subtract 1+x and divide by x^2 to get $1/2+O(x)$. The limit is 1/2.

Watch the trap: When a numerator's leading terms cancel, a lower-order approximation can erase the very term you need.

Recall task: Classify x^6 and x^5 at zero.

Worked answer: x^6 has a strict local (and global) minimum at zero. x^5 changes sign and has no extremum there, despite a zero derivative.

06 / Maxima, minima & optimization

Candidate points, global comparisons, modeling and iterative methods

Local tests and global extrema

A local minimum is no larger than nearby domain values; a global minimum is no larger than every value in the domain. A differentiable interior extremum must satisfy $f'=0$ (Fermat's theorem). The converse is false: x^3 has a stationary point without an extremum.

Global search on [a,b]: endpoints + interior $f'=0$ + interior nondifferentiable points

Work it through: For $f=x^3-3x$ on $[-2,2]$, $f'=3x^2-3$ vanishes at -1 and 1. Evaluate $f(-2)=-2$, $f(-1)=2$, $f(1)=-2$, $f(2)=2$. The global maximum is 2 at $x=-1$ and 2. The global minimum is -2 at $x=-2$ and 1.

Watch the trap: Solving $f'=0$ alone misses corners and boundary optima, and includes non-extrema.

Recall task: Find global extrema of $|x-1|$ on $[0,3]$.

Worked answer: Minimum 0 at the corner $x=1$; maximum 2 at $x=3$.

Turn a word problem into one variable

Begin an optimization problem by defining variables and units, writing the objective, and identifying constraints. Use constraints to eliminate extra variables, leaving a single-variable function on a clearly stated feasible interval. A mathematically optimal but infeasible value is not a solution.

f convex and $f'(x^*)=0 \Rightarrow x^*$ global minimum · strict convexity ⇒ at most one minimizer

Work it through: A rectangle has perimeter 20, so if width is x, height is $10-x$. Area $A(x)=x(10-x)$, with $0\le x\le 10$ allowing degenerate endpoints. $A'=10-2x$ gives $x=5$; $A''=-2$ confirms strict concavity. The maximum area is 25, attained by the square 5×5.

Watch the trap: A formula without its feasible domain is an incomplete optimization model.

Recall task: For $x>0$, minimize $x+9/x$.

Worked answer: Derivative $1-9/x^2$ vanishes at $x=3$. Second derivative $18/x^3>0$ and both boundaries diverge; the minimum is 6.

Gradient descent, Newton steps and stopping

Single-variable gradient descent uses $x_{k+1}=x_k-\eta f'(x_k)$ with step size $\eta>0$. It moves downhill locally, but a step that is too large can increase the objective or diverge. A small derivative is a stationarity signal, not proof of a global minimum.

GD: $x_{next}=x-\eta f'(x)$ · Newton: $x_{next}=x-f'(x)/f''(x)$

Work it through: For $f=(x-3)^2$, $f'=2(x-3)$. Choose $\eta=0.25$ and start $x_0=0$. The update is $x_{next}=0.5x+1.5$. The first iterates are 1.5, 2.25 and 2.625. The error halves each step; $\eta=1$ would instead produce a non-decaying oscillation.

Watch the trap: Newton's root method $x-f/f'$ solves $f=0$; Newton's optimization method $x-f'/f''$ seeks stationarity. They are different targets.

Recall task: For $f=2x^2$, find the interval of constant positive learning rates that converges to zero from arbitrary x_0.

Worked answer: Here $a=4$, so $0<\eta<1/2$.

07 / Supporting bridge: integration & gradients

Extra preparation for probability, least squares and machine learning

Integration recap: accumulation and area

This is supporting preparation, beyond the explicitly named single-variable DA calculus list. An antiderivative F satisfies $F'=f$. Indefinite integration gives a family $F+C$ because derivatives erase constants. For $p\neq-1$, $\int x^p dx=x^{p+1}/(p+1)+C$; $\int 1/x\,dx=\ln|x|+C$ on intervals avoiding zero.

$$\int_a^b f=F(b)-F(a)\cdot \int u\,dv=uv-\int v\,du\cdot \int 2xe^{x^2}dx=e^{x^2}+C$$

Work it through: Compute $\int_0^1 x^2dx$. An antiderivative is $x^3/3$. Evaluate at the bounds: $1/3-0=1/3$. A midpoint sum with n bins uses $x_i=(i+1/2)/n$ and width $1/n$. As the mesh shrinks, the sum approaches 1/3.

Watch the trap: An indefinite integral needs +C; a definite integral is a number with signed area, not necessarily positive.

Recall task: Compute $\int_0^1 2x\ e^{\wedge}\{x^2\}\ dx$.

Worked answer: Substitute $u=x^2$, $du=2x\,dx$. The result is $\int_0^1 e^u du=e-1$.

Partial derivatives, gradients and Hessians

This lesson is an application bridge rather than a separately named Section 3 topic. For $f(x_1,\ldots,x_n)$, a partial derivative changes one coordinate while holding the others fixed. The gradient collects the partial derivatives into a vector; its dot product with a unit direction gives the directional derivative when f is differentiable.

$$\nabla f=(\partial f/\partial x_i)\cdot Duf=\nabla f^T u\ (||u||=1)\cdot H_{ij}=\partial^2 f/(\partial x_i\partial x_j)$$

Work it through: For $f=x^2+xy+2y^2$, $\nabla f=(2x+y,x+4y)$. The Hessian is $[[2,1],[1,4]]$, symmetric with positive leading minors 2 and 7. The unique stationary point is (0,0). Positive definiteness proves a strict global minimum because the entire function is quadratic.

Watch the trap: Positive semidefinite Hessian at a single stationary point does not alone prove a minimum for a general nonlinear function.

Recall task: Find gradient and Hessian of x^2-y^2; classify the origin.

Worked answer: Gradient $(2x,-2y)$, Hessian $\mathrm{diag}(2,-2)$, so the origin is a saddle.

A capstone: fit a line and explain the loss

Combine the two tracks by fitting $y\approx\beta_0+\beta_1 t$ to observations. Build a design matrix X with a column of ones and a column of t values. The prediction vector $X\beta$ lies in the column space of X; least squares projects observed y onto that space.

$$\beta^*=\operatorname{argmin}\ \tfrac{1}{2}||X\beta-y||^2\cdot X^T r=0\cdot H=X^T X$$

Work it through: Use $t=(0,1,2)$, $y=(1,2,2)$. Then $X^TX=[[3,3],[3,5]]$ and $X^Ty=(5,6)$. Solve $3\beta_0+3\beta_1=5$ and $3\beta_0+5\beta_1=6$. $\beta_1=1/2$ and $\beta_0=7/6$; predictions are $(7/6,5/3,13/6)$. Residual $y-X\beta=(-1/6,1/3,-1/6)$; both its sum and t-weighted sum are zero.

Watch the trap: A prediction residual can be nonzero at the optimum; what vanishes is its projection onto every design-column direction.

Recall task: Fit a constant to observations 2,4,9 and give the residual vector.

Worked answer: The best constant is their mean 5. Residuals are $(-3,-1,4)$, summing to zero.

A final retrieval pass

1. f^{-1} denotes…

Inverse function / Reciprocal function always / Negative derivative / Second derivative

2. $\lim_{x\to 0} x \sin(1/x)$ equals…

Does not exist / 0 / 1 / ∞

3. Derivative of x^5 is…

$5x^4$ / x^4 / $5x^5$ / $x^6/6$

4. If $f''>0$ throughout an interval, the slope is…

Always zero / Increasing / Decreasing / Always positive

5. If $|f'''|\le M$, a degree-two Taylor error is bounded by…

$M|x-a|$ / $M/3$ / $M|x-a|^3/6$ / $M|x-a|^2/2$

6. A local minimum is necessarily global?

Yes for every differentiable function / Yes on every bounded domain / Yes if $f'=0$ / No

7. $\int_{-1}^{1} x\,dx$ equals…

2 / −1 / 0 / 1

8. For $f=x^2$, $g=x+1$, $f\circ g$ is…

x^2+1 / x^2+x / $2x+1$ / $(x+1)^2$

9. L'Hôpital applies directly to which form, with its other hypotheses?

0/0 / 1/0 / 2/3 / 0/1

10. Derivative of e^{2x} is…

e^{2x} / 2xe^{2x} / e^x / 2e^{2x}

Answers & reasoning

1. Inverse function

It reverses a one-to-one mapping.

2. 0

Squeeze its absolute value by $|x|$.

3. $5x^4$

Apply the power rule.

4. Increasing

The second derivative measures change in the first.

5. $M|x-a|^3/6$

Use the third derivative and 3! in the remainder.

6. No

Multiple wells can have different values; convexity provides a useful global guarantee.

7. 0

Signed areas cancel.

8. $(x+1)^2$

Apply g first and then square.

9. 0/0

A qualifying indeterminate quotient is needed.

10. 2e^{2x}

Chain rule multiplies by the inner derivative 2.